Colour Maximite 2

App User Guide

The programs that come with the Colour Maximite 2

For firmware 7.00.00RC3 and later

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The firmware, the programs and this guide are at
https://github.com/UKTailwind/CMM2V1.5

About This Guide

This guide is for anyone using the programs that come with the Colour Maximite 2: MicroPython, Doom, Heretic, Quake, ScummVM, the NES, ZX Spectrum and Amstrad CPC emulators, the 386 PC, the C compiler, the VT100 terminal and zeptoforth. It covers installing each one, starting it, its keys and leaving it. It assumes you can use the MMBasic command prompt and copy files to the SD card.

Support

Questions should be raised on the Back Shed forum (http://www.thebackshed.com/forum/Microcontrollers), and problems with the programs reported as issues of the repository above.

Copyright

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Introduction

Your Colour Maximite 2 can run much more than MMBasic. This guide covers the programs that come with the firmware: games, emulators, a C compiler, MicroPython, Forth and a terminal. It tells you how to install each one, start it, use it and leave it.

You do not need to be a programmer to use any of them, apart from the C compiler, MicroPython and zeptoforth. If you want to write programs of this kind yourself, that is what the companion manual *Colour Maximite 2 Loadable Programs* is for.

What Is Here

Program	What it does	Folder
MicroPython	Python, with the screen, sound and I/O pins	A:/
Doom	The classic game, with the free Freedoom game data	A:/Doom
Heretic	Raven Software's game, with its shareware episode	A:/heretic
Quake	id Software's game, with its shareware episode	A:/quake
ScummVM	Classic adventure games from LucasArts, Sierra and others	A:/quest
NES	A Nintendo Entertainment System emulator, with two free games	A:/nes
ZX Spectrum	A 48K and 128K Spectrum emulator	A:/spectrum
Amstrad CPC	A 464, 664 and 6128 emulator	A:/cpc
386 PC	A PC that runs FreeDOS, MS-DOS, Windows 3.1 and Windows 95	A:/pc386
C compiler	Compiles C and MMBasic programs on the machine itself	A:/cc
VT100 terminal	Makes the Colour Maximite 2 the screen and keyboard of a PicoMite or other computer	A:/vt100
zeptoforth	A Forth with tasks, double precision and a screen editor	A:/zeptoforth

All of them run on both the Generation 1 and the Generation 2, from the same files, except zeptoforth, which has a file for each, and Quake, which needs a Generation 2.

Installing the Programs

The programs are in the file CMM2_Programs_7.00.00RC3.zip, which is posted with the firmware.

1. Update the firmware first. Some of these programs need the firmware of the same release.

2. Unzip the file on a PC.

3. Copy everything in it to the root of the SD card, keeping the folders. Use a card reader: some of the files are tens of megabytes, which would take many minutes over the serial console.

You need only copy the folders you want. If you do not want Doom, leave out the Doom folder. Keep the licences folder with the programs if you pass them on to anyone else.

Starting a Program

A program is a file ending .app, and MMBasic runs it with RUN. Change to its folder first, as most of them look there for their files and keep their settings there:

```
CHDIR "A:/nes"
RUN "nes.app"
```

Words after a comma are passed to the program:

```
RUN "scummvm.app", dott
```

Do not put them in quotes. Each chapter of this guide says what its program accepts.

A BASIC program can start one too. MMBasic carries on with the next statement when it ends, with the program's variables as they were.

Leaving a Program

Programs differ in how you leave them, as an emulator has to pass most keys on to the machine it is emulating:

Program	How to leave it
MicroPython	machine.reset() at its prompt
Doom, Heretic, Quake	Ctrl-C
ScummVM	Ctrl-C. In a game, Ctrl+Alt+Del goes back to the list of games
NES	Ctrl+Alt+End, or Ctrl-C
ZX Spectrum, Amstrad CPC	Ctrl+Alt+End. Ctrl-C is a key of the emulated machine
386 PC	Ctrl+Alt+End. See its chapter first, as this switches the PC off
VT100 terminal	Shift-F12, then Quit. Ctrl-C goes to the computer at the other end

When a program ends, MMBasic tidies up after it: its files are closed, its sound is stopped, its I/O pins are released, and the video mode, font, colours and palette are put back.

What the Programs Use

The keyboard and mouse. All the programs use a USB keyboard. Those that use a mouse take a USB mouse, or a PS/2 mouse set up with OPTION MOUSE.

USB gamepads. The NES, Spectrum and CPC emulators use them. A DualShock 3 sends nothing until its PS button has been pressed. A generic pad is usually a low speed device, and has to go straight into the Colour Maximite 2, not through a hub.

The screen. A program sets the video mode it needs, and MMBasic puts yours back when it ends. Your monitor may take a moment to lock on each time.

Memory. A Generation 2 keeps 8MB of its memory for these programs. A Generation 1 lends a program the largest free part of MMBasic's memory while it runs, and some programs borrow the display memory too, so on a Generation 1 the screen can change as a program starts.

If a Program Will Not Start

Message	What it means
Needs a newer firmware	The program needs a later version of the firmware than you have. Update it.
Not a loadable program	The file is not a program, or is not the one you meant.
The program file is damaged	Copy it to the card again.
Needs a G2	The program needs a Generation 2. Of the programs in this guide, only Quake does.
Needs n KB, n KB free in one block	There is not enough memory. Restart the Colour Maximite 2 and try it from the command prompt, not from inside a BASIC program. On a Generation 1, OPTION RAM leaves less: use OPTION FLASH.

If a program cannot find its files, you have probably started it from the wrong folder. CHDIR to its folder and try again.

MicroPython

MicroPython is the Python language for small computers. This is the MicroPython of the Pico Computer 3, so programs written for that run here too.

Starting It

micropython.app goes in the root of the card. At the MMBasic prompt:

```
RUN "A:/micropython.app"
```

It runs boot.py and main.py if they are in the current folder, and then shows its >>> prompt. To run one program and go straight back to MMBasic, name it:

```
RUN "A:/micropython.app", myprog.py
```

Using It

The prompt appears on the screen, and on the serial console. Type Python at it:

```
>>> print("Hello", 6 * 7)
Hello 42
```

These are ready at the prompt as it starts:

Name	What it does
ls(), cd(), cat()	List a folder, change folder, show a file
edit()	The editor
run()	Run a program
fm()	The file manager
plot(), Turtle()	Drawing and turtle graphics
play(), tone()	Sound
draw_jpg()	Show a JPEG picture
keydown(), mouse()	Read the keyboard and the mouse

The Keys

Keys	What they do
Ctrl-C	Cancels a line, or stops a running program
Ctrl-D	A soft reset, which starts MicroPython again

To leave MicroPython and go back to MMBasic, type:

```
import machine
machine.reset()
```

Your Files

The SD card is at / and also at /sd. A USB flash drive is at /usb when one is plugged in. Modules of your own can go in /lib.

What It Has

- The Pico Computer 3's modules for the display, the keyboard, the mouse, sound and pictures.
- SQLite, the database, and ulab, for arrays and fast numerical work.
- The machine module, for the rear I/O pins, by the PicoMite's GP numbers.

Python runs about twice as fast here as it does on the RP2350.

Doom and Heretic

Both are the original games, running at full speed on either generation.

Doom

```
CHDIR "A:/Doom"
RUN "doom.app"
```

This plays Freedoom Phase 1, a free replacement for Doom's game data. The music is played on the firmware's FM synthesiser.

Always start it from its folder, so that its settings and saved games are kept there.

Your Own Copy of Doom

If you have Doom's own game data, there are two ways to use it:

- Copy it into A:/Doom as doom.wad or doom1.wad, and remove freedoom1.wad.
- Name it on RUN's line:

```
RUN "doom.app", -iwad x.wad
```

Heretic

```
CHDIR "A:/heretic"
RUN "heretic.app"
```

The folder holds the whole of Heretic's shareware version 1.2, the first episode, unchanged. HERETIC1.WAD is what the game plays, and the DOS programs beside it are not used here.

If you have the full game, put its heretic.wad in the folder and it is used in place of the shareware.

LICENSE.DOC, in the same folder, says how the shareware may be passed on: free of charge, and compressed if it is sent electronically.

The Keys

Action	Doom	Heretic
Move and turn	The arrows	The arrows
Fire	Ctrl	Ctrl
Open, use	Space	Space
Run	Shift	Shift
Strafe	Alt with the arrows, or , and .	Alt with the arrows, or , and .
The menu	Esc	Esc
Choose an artifact		[and]
Use an artifact		Enter
Leave the game	Ctrl-C	Ctrl-C

Saving and loading are in the game's own menu, under Esc.

Quake

This is id Software's Quake, with its shareware episode. It needs a Generation 2. It draws 320 by 240 in 256 colours, and its timed demo runs at about 53 frames a second with the sound on.

```
CHDIR "A:/quake"
RUN "quake.app"
```

Always start it from its folder. It finds its game data there, and keeps its settings and saved games in the id1 folder inside it.

The folder holds the whole of Quake's shareware version 1.06, unchanged. ID1/PAK0.PAK is what the game plays, and the DOS and Windows programs beside it are not used here. SLICNSE.TXT, in the same folder, says how the shareware may be passed on: free of charge, as a whole, and compressed if it is sent electronically.

Sound effects play, at PLAY VOLUME's level. Quake's music was on its CD, so there is none here, and there is no network play.

The Keys

Action	Key
Move forward and back	Up and Down
Turn	Left and Right
Sidestep	, and . or Alt with Left and Right
Run	Shift
Fire	Ctrl, or the left mouse button
Jump, or swim up	Space or Enter
Look up and down	A and Z
Look straight ahead	End
Choose a weapon	1 to 8, or / for the next one
Show the scores	Tab
The menu	Esc
Help	F1
Save and load	F2 and F3
Quick save and quick load	F6 and F9
The console	The key to the left of 1, called the backquote
Leave the game	F10 then Y, or Ctrl-C

The keys can be changed under Options in the menu.

The Mouse

Moving the mouse turns you. The left button fires, and the right button walks forward. Hold the middle button to look up and down with the mouse as well.

To keep looking with the mouse without holding a button, open the console and type:

```
+mlook
```

The sensitivity is under Options in the menu.

The Mouse Wheel

Quake leaves the wheel unused, and turning it shows a message such as "MWHEELUP is unbound, hit F4 to set". To make it change weapons, open the console and type:

```
bind mwheelup "impulse 10"
bind mwheeldown "impulse 12"
```

Settings

Settings and key changes are saved in id1/config.cfg when you leave the game with F10 or Ctrl-C. If the Colour Maximite 2 is restarted while the game is running, any changes since it started are lost.

Mouse look is not one of the settings saved. To have it on every time, make a text file called autoexec.cfg in A:/quake/id1, which the game runs each time it starts:

```
+mlook
bind mwheelup "impulse 10"
bind mwheeldown "impulse 12"
```

Your Own Copy of Quake

If you have the full game, copy its PAK0.PAK and PAK1.PAK into A:/quake/id1 in place of the shareware's PAK0.PAK. This has not been tried: the program has only been tested with the shareware.

The game has 12MB of MMBasic's memory. If it runs out, ask for more on RUN's line, up to about 15:

```
RUN "quake.app", -mem 15
```

ScummVM

ScummVM plays classic adventure games from their original data files. This version comes from FRANK Quest, rh1tech's port of ScummVM to the RP2350. It has the AGI, SCI, SCUMM (versions 1 to 7), Gobliiins and Kyrandia engines.

Starting It

```
CHDIR "A:/quest"
RUN "scummvm.app"
```

It lists the games it finds in folders of A:/quest. Choose one with the arrow keys and press Enter.

To start a game at once, give its folder's name:

```
CHDIR "A:/quest"
RUN "scummvm.app", dott
```

On a Generation 1 the screen changes to MODE 7 as it starts, as ScummVM borrows the display memory.

The Keys

Keys	What they do
The mouse	Most of these games are played with it
F5	Save and restore (see below)
Esc	Skips a cutscene, in most games
Ctrl+Alt+Del	Back to the list of games
Ctrl-C	Leave ScummVM

Saving Your Game

Saved games are kept in A:/quest/saves. How you save depends on who made the game.

LucasArts Games

This means Monkey Island, Indiana Jones, Day of the Tentacle, Sam & Max, Loom and Full Throttle.

Press **F5** to open the save menu. It lists nine slots, with the date each one was saved.

Keys	What they do
Up, Down or 1 to 9	Choose a slot
S	Save in that slot. If the slot is in use it asks first
R	Restore that slot
Esc or F5	Close the menu

A message tells you when the game has been saved.

There are also shortcuts, which work without the menu and use the same slots:

Keys	What they do
Alt-1 to Alt-9	Save in that slot at once, replacing what was there
Ctrl-1 to Ctrl-9	Restore that slot at once

You cannot save during a cutscene, or before the game has properly started. The menu says so if that is why.

Sierra Games

The AGI and SCI games use their own save and restore screens, as they did on a PC. Press **F5** to save and **F7** to restore, or use the game's menu bar, which Esc opens in most of them. You can choose a slot and give each save a name.

Gobliiins and Kyrandia

These have save and restore in their own menus, on the game's screen.

The Games That Come With It

Folder	Game
freddy	Freddy Pharkas, Frontier Pharmacist: the interactive demo (Sierra, 1993)
agi	DG: The Adventure Game (Robin Gravel, 2001; public domain)
dgc64	Another DG Game: I Want My C64 Back (Robin Gravel, 2002; public domain)

Adding Games You Own

Games you own go in folders of A:/quest. The folder's name tells ScummVM which game it is, so it must be one of those below, in upper or lower case. A folder with any other name is not listed.

Copy the game's data files only, not its .EXE and .DRV files.

LucasArts (SCUMM)

Folder	Game
mi1	The Secret of Monkey Island, VGA floppy (needs 000.LFL)
mi1ega	The Secret of Monkey Island, EGA floppy (needs 000.LFL)
monkey, mi1cd	The Secret of Monkey Island, CD (needs monkey.000)
monkey2, mi2	Monkey Island 2: LeChuck's Revenge (needs monkey2.000)
loom	Loom, VGA (needs 000.LFL)
indy4, atlantis	Indiana Jones and the Fate of Atlantis (needs atlantis.000)
dott, tentacle	Day of the Tentacle (needs tentacle.000)
samnmax, sam	Sam & Max Hit the Road
ft, fulltp	Full Throttle (needs ft.la0)

Sierra (SCI)

Folder	Game
kq1 kq4 kq5 kq6	King's Quest I, IV, V, VI
sq1 sq3 sq4 sq5	Space Quest I, III, IV, V
lsl1 lsl2 lsl3 lsl5 lsl6	Leisure Suit Larry 1, 2, 3, 5, 6
pq1 pq2 pq3	Police Quest I, II, III
qfg1 qfg1vga qfg2 qfg3	Quest for Glory I (EGA, VGA), II, III
laurabow laurabow2	Laura Bow 1 and 2
ecoquest ecoquest2	EcoQuest 1 and 2
iceman longbow camelot	Codename: ICEMAN, Conquests of the Longbow, of Camelot
freddy jones pepper slater	Freddy Pharkas, Jones in the Fast Lane, Pepper's Adventures in Time, Slater & Charlie
castlebrain islandbrain mothergoose	Castle and Island of Dr. Brain, Mixed-Up Mother Goose

Others

Folder	Game
kyra1, kyrandia	The Legend of Kyrandia (needs KYRA.DAT: copy it from A:/quest into the game's folder)
kyra2, handoffate	The Legend of Kyrandia 2: The Hand of Fate (needs KYRA.DAT too)
gob1 gob2 gob3	Gobliiins, Gobliins 2, Goblins 3
agi	One AGI game (version 2) of any kind, such as a fan game

Other games, and demos other than those of Gobliiins and Kyrandia, are not supported. Nor are LucasArts' last two: The Dig (SCUMM version 7, not built into FRANK Quest) and The Curse of Monkey Island (SCUMM version 8, at 640 x 480).

The NES Emulator

This is frank-nes, Mikhail Matveev's NES emulator, on the QuickNES core by Shay Green. It runs at full speed on both generations, in MODE 7,8.

Starting It

```
CHDIR "A:/nes"
RUN "nes.app"
```

Its cartridges are the .nes files in A:/nes. It comes with two free games, Nova the Squirrel and Thwaite. To add more, copy the .nes files into the folder.

The welcome screen moves on by itself after 10 seconds, whether or not a button has worked.

The Controls

It uses USB gamepads and the USB keyboard.

Pad	Keyboard	NES
D-pad or stick	The arrows or WASD	D-pad
Circle, Triangle	X	A
Cross, Square	Z	B
Start	Enter	Start
Select	Space	Select

These work at any time:

Keys	What they do
Start and Select together, or F12	The menu
F11	Back to the cartridges
Ctrl+Alt+End, or Ctrl-C	Back to MMBasic

If a Gamepad Does Nothing

- A DualShock 3 sends nothing until its PS button has been pressed.
- A generic pad is usually a low speed device, and has to go straight into the Colour Maximite 2, not through a hub.
- If a generic pad gives no response after a restart, unplug it and plug it in again.

The ZX Spectrum Emulator

This is pico-spec, DnCraptor's port of the ESPectrum emulator by Victor Iborra and David Crespo. It runs at the Spectrum's 50 frames a second on both generations, in MODE 7,8.

Starting It

```
CHDIR "A:/spectrum"
RUN "spectrum.app"
```

It starts as a 48K Spectrum. The ROMs are built into the program, so it needs nothing else on the card. The F1 menu chooses another machine. The 48K and 128K have been tried here; the others in its menu are as pico-spec has them.

Loading Games

Put tapes and snapshots in A:/spectrum, and open them with **F5**, which starts a tape by itself. You can also choose a tape and type LOAD "" as you would on a real Spectrum.

The program calls the folder it was started in SD:/, and keeps its settings there.

The Keys

Keys	What they do
F1	The menu
F3, F4	Load and save a snapshot, in numbered slots
F5	Open a file
F6	Start or stop the tape
F9, F10	Volume down and up
F11	Reset the Spectrum
F12	Restart the emulator. It asks first, with No chosen
Shift	Caps Shift
Ctrl	Symbol Shift
Ctrl+Alt+End	Back to MMBasic

Ctrl-C is a Spectrum key here, Symbol Shift and C, so it does not end the program.

Snapshots are how you save your place in a game: F4 saves one and F3 loads it.

USB gamepads step through the emulator's menus, and are the Spectrum's joysticks. They are Kempston joysticks unless the menu says otherwise.

The Amstrad CPC Emulator

This is frank-cpc, rh1tech's port of the Caprice32 emulator. It runs at the CPC's 50 frames a second on both generations, in MODE 8,8.

Starting It

```
CHDIR "A:/cpc"
RUN "cpc.app"
```

It starts as a CPC 6128. Press a key to clear its welcome screen. **F12** chooses the 464, 664 or 6128, and 64K, 128K or 576K of memory. Its ROMs are in its rom folder.

The program calls the folder it was started in /cpc, and keeps its settings there in cpc.ini.

Loading Games

Put disks (.dsk), tapes (.cdt, .cas) and cartridges (.cpr) in A:/cpc/disk, and open them with **F11**.

- For a tape on the 664 or 6128, type |TAPE first.
- Leave Fast Tape set to Off.

Saving your work. A SAVE to a disk is lost when the disk is ejected or the program ends. Use a snapshot, in the F12 menu.

The Keys

Keys	What they do
F11	Disks, tapes and cartridges
F12	The settings
Ctrl+Alt+Del	Reset the CPC
Ctrl+Alt+End	Back to MMBasic
Backspace	DEL
Delete	CLR
Ctrl	CONTROL
Alt	COPY
Caps Lock	CAPS LOCK
Keypad Enter	The small ENTER

The other keys give what is printed on them, following OPTION USBKEYBOARD, so the symbols are where your keyboard has them.

Ctrl-C is a CPC key here, so it does not end the program. USB gamepads are the CPC's joystick.

The 386 PC

pc386 is tiny386, Chunhui He's 386 PC emulator, with a VGA card, a Sound Blaster 16, an Adlib and the PC speaker. The Colour Maximite 2's keyboard and mouse are the PC's. It runs FreeDOS, MS-DOS, Windows 3.1 and Windows 95.

This chapter gets you started. Everything else, including installing MS-DOS and Windows, is in its own manual, *Colour Maximite 2 386 PC Emulator*, posted with the firmware.

Starting It

```
CHDIR "A:/pc386"
RUN "pc386.app"
```

The screen changes to 640 x 480, and FreeDOS 1.3 starts from a 32 MB hard disk image. It loads its mouse driver and ends at the C:\> prompt.

A PC is described by a configuration file. The one used unless you say otherwise is pc386.ini. To start a different PC, name its file:

```
RUN "pc386.app", msdos.ini
```

msdos.ini installs MS-DOS 6.2 from images of your own Setup disks, and msdosc.ini then starts it, with Windows 3.1's disks to hand.

Leaving It

Press **Ctrl+Alt+End**. This switches the PC off at once, as if its power had been cut, so first:

- Finish what the PC is doing, and leave Windows if it is running.
- Wait until the PC has stopped writing to its disks.

The PC's disks are image files on the SD card, so everything written to them is kept for the next time.

The Keys

Three keys, held with Ctrl and Alt, are the emulator's own. Every other key goes to the PC, Ctrl+Alt+Delete included: it restarts the PC. Ctrl-C goes to the PC as well, and does not stop the emulator.

Keys	What they do
Ctrl+Alt+End	Leave the emulator
Ctrl+Alt+Page Down	Put the next floppy image of the list in drive A:
Ctrl+Alt+Page Up	Put the previous floppy image in drive A:

The keypad's 1, 3 and 9 work too, if a keyboard does not report those combinations.

DOS uses a US keyboard layout unless you load another. For a UK keyboard, put this in AUTOEXEC.BAT:

```
KEYB UK
```

The Mouse and Sound

DOS programs need a mouse driver. The FreeDOS disk loads CuteMouse, and Windows 3.1 has its own.

Sound comes out of the Colour Maximite 2's audio output. DOS programs find the Sound Blaster at port 220, IRQ 5, DMA 1 (high DMA 5), and an Adlib at port 388. Many of them read these settings from the BLASTER variable, which AUTOEXEC.BAT can set:

```
SET BLASTER=A220 I5 D1 H5 T6
```

Making It Faster

- OPTION SD TIMING FAST makes the SD card, and so the PC's disks, much faster.
- On an original Colour Maximite 2 whose processor is of revision Y, OPTION OVERCLOCK ON makes it as fast as on a Generation 2.

Moving a PC

A PC is a few files in a folder, so copying them moves it to another Colour Maximite 2 of either generation. FreeDOS can be passed on to other people. MS-DOS, Windows and other commercial programs cannot.

The C Compiler

The Colour Maximite 2 can compile programs on the machine itself, with no PC. The CC command compiles a program written in C. It also compiles an MMBasic program, so that a program that spends its time calculating runs much faster.

This chapter covers everyday use. The full details are in its own manual, *Colour Maximite 2 C Compiler*, posted with the firmware.

Installing It

The cc folder has to be at A:/cc. CC looks for it there, whatever the current folder is. It needs firmware 7.00.00RC2 or later.

File	What it is
A:/cc/cc.app	The compiler
A:/cc/include	The header files that C programs include
A:/cc/examples	Two example programs, shapes.c and blink.c

A compiled program needs nothing from A:/cc to run, so you can pass it on by itself.

When It Is Worth Using

Your program	What to use
An MMBasic program that is fast enough	Nothing: leave it as it is
An MMBasic program that spends its time calculating	CC, which is much the fastest way to run it
An MMBasic game that spends its time drawing	Try OPTION COMPILE ON first. CC gains less here, as the drawing takes the same time however the program is run
Something new that has to be as fast as it can be	Write it in C

Compiling an MMBasic Program

Your BASIC program is not changed. RUN "game.bas" still runs it with the interpreter, and the compiled copy sits beside it in a file of its own. Change to the program's folder first, as a game finds its pictures and sounds from there:

```
CHDIR "A:/game"
CC "game.bas"
RUN "game.b.bc"
```

CC prints a row of dots as it works. A program of about 3,000 lines compiles in 7 seconds. The largest tried so far, 4,300 lines, takes about a minute and a half, and needs a Generation 2.

If you change game.bas, run CC again. RUN "game.b.bc" does not check that it is up to date.

Read What CC Reports

If CC cannot translate a statement it does not stop. **The statement is left out**, and CC reports the line and the reason. Nothing runs in its place, so a program with a line left out may run and do the wrong thing. Read the report, particularly if the line left out opened a block.

Check the Result

A compiled program should behave as the original does, but yours may do something that has not been tried. Run the BASIC program and the compiled one, and check that they agree before you rely on it.

A compiled program is faster, so a game that paces itself by how long its own code takes will run too fast. It needs a timer in its main loop.

Compiling a C Program

Type this into a file called hello.c, with EDIT or on a PC:

```
#include <stdio.h>

int main(void)
{
    printf("Hello from C, %d\n", 6 * 7);
    return 0;
}
```

Then, in the same folder:

```
CC "hello.c"
RUN "hello.bc"
```

CC makes hello.bc beside hello.c, in about a second. MMBasic carries on when the program ends, and Ctrl-C stops a program as it does in BASIC.

RUN can do both steps. RUN "hello.c" compiles the file if hello.bc is missing or older than it, and then runs it, so editing a program and trying it again is one step.

Words after a comma are the program's arguments, in argv[1] onward:

```
RUN "hello.bc", one two
```

The Language

The language is C89 (ANSI C) with long long, // comments and declarations after statements. A program is one source file, as there is no linker. Bit fields are not supported, nor are declarations inside a for statement.

The usual stdio.h, stdlib.h, string.h, math.h, ctype.h and time.h are there. There is no scanf. Files have MMBasic's names, such as A:/dir/name.txt.

cmm2.h gives a C program the Colour Maximite 2's own facilities: the video modes, drawing with MMBasic's own routines, the keyboard, the mouse, sound and the pins:

```
#include <cmm2.h>

int main(void)
{
    cmm2_video v;

    video_mode(1, 8, &v);
    gfx_cls(BLACK);
    gfx_circle(400, 300, 100, 2, 1.0, GREEN, RED);
    gfx_text(400, 300, "Hello", "CM", 1, 2, WHITE, NOFILL);
    return 0;
}
```

CC's Options

CC takes a second argument, a string of options:

Option	What it does
-v	Shows each pass of the compiler as it runs
-k	Keeps the intermediate files
-o name	Gives the object file another name

For example:

```
CC "hello.c", "-o greet.bc"
```

How Much Faster

A 3,200-line calculation, on a Generation 2 at 480MHz:

How it ran	Time	Compared with the interpreter
The interpreter	6.16 seconds	1
OPTION COMPILE ON	1.71 seconds	3.6 times faster
CC	0.93 seconds	6.6 times faster

Games gain less, as they spend much of their time drawing. Try OPTION COMPILE ON first: it needs no extra step and works on any program.

When Things Go Wrong

What you see	What to do
CC cannot find the compiler	cc.app has to be in A:/cc. Copy the cc folder from the zip to the card, and check its name
Cannot open include file stdint.h	The files in A:/cc/include are from an older version. Copy the whole cc folder from the zip that came with your firmware
x.c did not compile	The lines before it give the file, line number and error. A missing semicolon is reported on the next line
A compiled BASIC program does the wrong thing	Look at what CC reported. A line it could not translate is left out
A game runs too fast	It needs a timer in its main loop, as the compiled program is faster
not enough memory for its tables, jump out of range, code segment full	The program is too large for the compiler on a Generation 1. Compile it on a Generation 2
The compiled program cannot find its files	CHDIR to the program's folder before RUN

The VT100 Terminal

This program makes the Colour Maximite 2 a terminal: the screen and keyboard of another computer. Plug a PicoMite into it and you can use the PicoMite's prompt, its editor and its file manager, in colour, with the mouse, without a PC.

It has a setup menu, scrollback, a log, and XMODEM for moving files. It is a VT100 and VT102, with xterm's 256 colours and mouse reports added, and VT52 mode. It uses libvterm, with the Spleen fonts.

It needs firmware 7.00.00RC3 or later for USB serial devices.

Connecting

Port	What it is
COM4 to COM7	USB serial devices, such as a PicoMite or a USB serial adapter, in a USB socket or through a hub
COM1	The serial port on GP0 and GP1 of the rear connector
COM2	The serial port on GP36 and GP37 of the rear connector

The rear connector's serial ports use 3.3 V signals and are not 5 V tolerant. Do not connect them to an RS-232 port.

When a USB serial device is plugged in, the MMBasic prompt shows which port it is, starting at COM4.

Starting It

```
CHDIR "A:/vt100"
RUN "vt100.app", COM4 115200
```

The port and speed can be left out once they are saved in the setup menu. Any setting can be given on RUN's line by name, such as font=3.

The line at the bottom of the screen shows the port, its speed and the size of the terminal.

Leaving It

Press **Shift-F12**, or Pause, to open the setup menu, then choose **Quit**.

Ctrl-C does not leave the terminal. It goes to the computer at the other end, as it should, to stop a program there.

Using It With a PicoMite

Everything works as it does on a terminal on a PC:

- The PicoMite's prompt, EDIT and FM work with the arrow keys, the function keys and the mouse.
- When the PicoMite asks for a screen size, at EDIT, FM and OPTION DISPLAY, the terminal changes to it. It tries a smaller font in the same video mode first, so your monitor rarely has to lock on again.
- Colours are shown as the PicoMite sends them.

The block on the screen in EDIT is the mouse pointer, and the underline is the text cursor.

If the PicoMite Restarts

Many PicoMite OPTION commands restart it, and it also drops off when you unplug it. You do not have to do anything: the status line says **COM4 disconnected, waiting for it to come back**, and when the device returns the terminal carries on by itself.

The Keys

Keys	What they do
Shift-F12, or Pause	The setup menu
Shift with Page Up, Page Down	Look back through what has scrolled off the screen
Shift with Home, End	The top of the scrollback, and back to the live screen
The mouse wheel	Scrollback too, unless the other computer is using the mouse
Any other key	Goes back to the live screen, and to the other computer
Ctrl-J	Sends a line feed, on its own

The Setup Menu

Up and Down choose a setting, and Left, Right or Enter change it. Esc goes back to the terminal. Choose **Save settings** to keep your changes, which go in vt100.cfg in the current folder.

Setting	What it does
Port, Speed, Format	The connection. Format is the data bits, parity and stop bits
Screen	The video mode
Font	From 5 x 8 to 16 x 32. 8 x 14 is the default
Columns, Rows	The terminal's size. All uses the whole screen
Status line	The line at the bottom
Characters	How bytes above 127 are shown: CP437 (as a PicoMite has them), Latin-1 or UTF-8
Text colour	White, green or amber, like a monitor of the time
Cursor	Underline, Block or Bar
Backspace sends	BS (8) or DEL (127)
Enter sends	CR, or CR LF. A PicoMite wants CR
LF received as	LF, or CR LF. This is for what arrives, and does not change what your keys send
Local echo	Shows what you type, for a computer that does not send it back
Text line delay	The pause after each line when sending a text file
Send text file...	Sends a file as if you had typed it
XMODEM send..., XMODEM receive...	Moving files (below)
Log to file...	Records everything that arrives in a file
Reset the terminal	Puts the terminal back as it starts
Save settings	Keeps these settings
Quit	Leaves the terminal

Moving Files With XMODEM

Files can be moved each way, between the SD card and a PicoMite or other computer.

From the Colour Maximite 2 to a PicoMite

1. At the PicoMite's prompt, type XMODEM RECEIVE "name.txt" and press Enter.
2. Straight away press Shift-F12, choose **XMODEM send...** and press Enter.
3. Pick the file from the list and press Enter.

The PicoMite waits only a limited time, so go straight on from step 1 to step 2.

From a PicoMite to the Colour Maximite 2

1. At the PicoMite's prompt, type XMODEM SEND "name.txt" and press Enter.
2. Press Shift-F12, choose **XMODEM receive...** and press Enter.
3. Type a name for the file and press Enter.

Esc stops a transfer. Retries in the progress display mean the line is having trouble, even if the file arrives whole.

File Sizes

XMODEM sends files in blocks of 128 bytes with no length, so the last block is padded. A text file arrives exactly as it was. A binary file keeps the padding, and can arrive up to 127 bytes longer. Nothing is lost.

zeptoforth

zeptoforth is Travis Bemann's Forth for ARM microcontrollers, at https://github.com/tabemann/zeptoforth. On the Colour Maximite 2 it is a complete Forth system with preemptive tasks, double-precision arithmetic, a screen text editor, and words for the screen, sound, files and I/O pins.

Why Forth

Forth is a very small language that you extend as you go. A program is a set of words. You build each word from the words you already have, and from then on it is part of the language, no different from the words that came with it. Numbers pass between words on a stack, so a word needs no declarations and no argument lists:

```
: square  dup * ;
7 square .
```

The first line defines a word that squares a number. The second puts 7 on the stack, squares it, and prints 49.

Forth is compiled and interactive at once. Each word becomes ARM machine code the moment you finish typing it, on the Colour Maximite 2 itself, with no PC and no build step, and you can try it at the prompt straight away. You build a program from the bottom up, testing each small word as you go, so a fault is usually in the last thing you typed.

The language is yours to change. The compiler is made of words too, so you can add control structures of your own or change how numbers are read. And nothing stands between you and the machine: words read and write any address, and there is an assembler when you need one.

Starting It

Copy the zeptoforth folder to the card, then on a Generation 2:

```
CHDIR "A:/zeptoforth"
RUN "zeptoforth.app"
```

On a Generation 1:

```
CHDIR "A:/zeptoforth"
RUN "zeptoforth_g1.app"
```

Each has its own saved Forth system, zeptoforth.img or zeptoforth_g1.img, in the same folder. Keep them together.

Leaving It

Type **bye** to go back to MMBasic.

Ctrl-C stops a word that is running and starts Forth again. What you compiled to RAM is lost, and what you saved is kept. Ctrl-C does not leave zeptoforth.

The Console

The console is a terminal on the screen, the same one as the VT100 terminal program, so Forth's colours and editors work on the Colour Maximite 2 with no PC. It is also sent to the serial port, and you can type at either.

Shift with Page Up and Page Down, or the mouse wheel, looks back through what has scrolled off the screen.

The font suits the video mode. To change it, or the size of the console:

```
cmm2 import
4 console-font
80 24 console-size
```

console-font takes 0 to 5, from 5 x 8 to 16 x 32, or -1 to go back to the one that suits the mode. console-size takes the columns and lines, and 0 0 means all there is.

To recall and edit what you have typed with the arrow keys, type **enable-line**. Up and Down go through the lines you typed before. disable-line turns it off, as you should before sending a file to the console from a PC.

Programs that ask how big the terminal is get the size of the screen, so a terminal program on a PC should be set to the same size, or use console-size.

Editing Files

zeptoed is a full-screen editor for files on the SD card or USB drive:

```
s" A:/zeptoforth/mycode.fs" zed
```

The file is made if it is not there. Names are MMBasic's, with the drive and folders, or relative to the current folder.

Key	What it does
Ctrl-W	Saves the file
Ctrl-V	Leaves the editor
Ctrl-Z	Undoes the last change
Ctrl-R	Searches forward
Ctrl-Space	Starts or clears a selection
Ctrl-K	Cuts the selection
Ctrl-Y	Pastes
Ctrl-O	Opens another file
Ctrl-N, Ctrl-P	Moves between the files open, and the help
Ctrl-? or Ctrl-/	Shows all the keys. Ctrl-P goes back to your file, and Esc then Ctrl-X closes the help

Files in the editor are held in 512 KB of memory, with what it needs to undo your changes. To change it, type for example 1000000 to zeptoed-heap-size.

Ctrl-C restarts Forth from the editor as it does anywhere, so save first.

Loading a Program

```
include mycode.fs
```

A file may include another, up to four deep. fat32-tools import gives zeptoforth's own file words, such as list-dir, list-file and copy-file. The module cmm2 has MMBasic's, such as files and chdir.

Keeping Your Work

Words go to RAM to begin with, and are lost at a restart. To keep them, compile them to the flash dictionary and save it:

```
compile-to-flash
include mycode.fs
compile-to-ram
save-image
```

save-image writes the whole Forth system back to zeptoforth.img or zeptoforth_g1.img, and it is there at the next start. What zeptoforth calls flash is memory here, so nothing is written to the Colour Maximite 2's own flash.

Keep a copy of the image file before you save over it. If a saved image stops zeptoforth from starting, put the copy back.

Blocks

Forth's blocks are kept in the file blocks.dat in the current folder, up to 4096 blocks of 1024 bytes. **n edit** opens block n in the block editor, Ctrl-V leaves it, and **n list** and **n load** work as they do in any Forth.

Graphics, Sound and I/O

The words for the screen, sound, the keyboard and mouse, files and the I/O pins are in the module cmm2:

```
cmm2 import
1 8 mode
300 300 80 2 1de0 cyan orange circle
```

WORDS.md in the folder lists them. Their arguments are in MMBasic's order. The console goes on working in any video mode, on top of what you draw. A graphics program should usually turn it off the screen with 0 console-screen, so that nothing it prints lands on the picture.

Double Precision

Decimal numbers have a d before the exponent, and the words have dv in front:

```
float64 import
2de0 dvsqrt dv.
```

The words are those of zeptoforth's single-precision float32, with v changed to dv: dv+, dvsin, dv. and so on. A value takes two cells. A program written with the names v+ and numbers like 1.5e0 can type include float64_v.fs and float64-v import.

Learning More

WORDS.md in the folder describes every word this version adds: the screen, sound, files and I/O, double precision, blocks and the rest. zeptoforth's own reference has everything else, at https://github.com/tabemann/zeptoforth/tree/master/docs/words.

If Forth is new to you, these are good places to start:

Starting Forth	**Leo Brodie's classic introduction, free at https://www.forth.com/starting-forth/**
Easy Forth	A short tutorial that runs in your browser, at https://skilldrick.github.io/easyforth/
Thinking Forth	Leo Brodie's book on how to design programs in Forth, free at https://thinking-forth.sourceforge.net

Forths differ in their details, so a word in a book may have another name here, or need a module imported.

Errors

A bad address gives the message **bad address**, and you go back to the prompt with what is in RAM kept. Inside a firmware service it restarts Forth instead.

If a Generation 1 says **the memory it is linked for is not free**, MMBasic is using too much of its memory. Run zeptoforth from the prompt after a restart, with no program running.

Memory

	Generation 2	**Generation 1**
RAM dictionary and stacks	3 MB	1 MB
Flash dictionary	2 MB	768 KB

The Forth system itself takes about 510 KB of the flash dictionary.

Appendix A: Licences

The programs have their own licences, which are in the licences folder of CMM2_Programs_7.00.00RC3.zip. README.txt in the same zip gives each in full, with the permissions for the ROMs. Keep the licences folder with the programs if you pass them on.

The source of each program is named in the file make.py of its folder in the firmware's repository, https://github.com/UKTailwind/CMM2V1.5, with the changes made for the Colour Maximite 2 beside it.

Program	Licence
MicroPython	MicroPython, ulab and usqlite: MIT. SQLite: public domain
Doom	GNU GPL version 2. Freedoom: BSD 3-Clause
Heretic	GNU GPL version 2. The shareware game data is copyright Raven Software and id Software, under its own licence (LICENSE.DOC): it is not free software and may not be sold
Quake	GNU GPL version 2. The shareware game data is copyright id Software, under its own licence (SLICNSE.TXT): it is not free software and may not be sold
ScummVM	GNU GPL version 2 or later, with parts under other licences as ScummVM's files say
NES	frank-nes: GNU GPL version 3. QuickNES: GNU LGPL version 2.1 or later. Nova the Squirrel: code GNU GPL version 3 or later, graphics and levels Creative Commons BY-NC-SA 4.0. Thwaite: GNU GPL version 3 or later
ZX Spectrum	pico-spec: GNU GPL version 3. miniz and the WD1793 emulation in it: MIT. The ROMs are not under the GPL (see README.txt)
Amstrad CPC	frank-cpc: GNU GPL version 3, with Caprice32 and cpc4x under GNU GPL version 2 or later. The ROMs are not under the GPL (see README.txt)
386 PC	tiny386: BSD 3-Clause. Its devices: MIT. inih: BSD 3-Clause. fmopl: GNU LGPL version 2.1 or later. SeaBIOS and SeaVGABIOS: GNU LGPL version 3. FreeDOS: each program under its own licence, most of them the GNU GPL
C compiler	The Fuzix Compiler Kit: GNU GPL version 3. The translator from MMBasic to C: the MMBasic licence
VT100 terminal	libvterm: MIT. The Spleen fonts: BSD 2-Clause
zeptoforth	zeptoforth: MIT. Its console has libvterm: MIT, and the Spleen fonts: BSD 2-Clause